

INTERNSHIP REPORT

Prompt Engineering and Generative AI

Submitted by

Prajwal Pratap Singh

Internship Organization

Devasthali Infotech Private Limited

Noida, Uttar Pradesh

Internship Duration

15 June 2026 – 15 September 2026

Date of Certificate Issue: 15 September 2026

Source Note

This report is prepared primarily from the internship certificate supplied for the internship. The certificate confirms the internship period, organization, location, and the nature of the work as exploration, testing, and refinement of text prompts for Generative AI tools to support business use cases. Where the report discusses general prompt-engineering concepts, evaluation methods, or illustrative examples, those sections are presented as technical background or examples rather than as claims that a specific tool, project, or client was used during the internship.

Intern	Prajwal Pratap Singh
Organization	Devasthali Infotech Private Limited
Internship Period	15 June 2026 to 15 September 2026
Primary Work Area	Exploration, testing, and refinement of text prompts for Generative AI tools
Purpose	Support business use cases
Place	Noida
Certificate Issue Date	17 September 2026

Certificate

The internship certificate states that Prajwal Pratap Singh successfully completed an internship with Devasthali Infotech Private Limited from 15 June 2026 to 15 September 2026. It also records contributions involving the exploration, testing, and refinement of text prompts for Generative AI tools in support of business use cases. The certificate was issued on 17 September 2026 in Noida and is signed by Rishi Kumar, Accounts Manager.



TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Prajwal Pratap Singh**, has successfully completed an internship with **M/s Devasthali Infotech Private Limited** from **15th June 2026** to **15th September 2026**.

During this period, he demonstrated enthusiasm, creativity and a strong willingness to learn, contributing to the exploration, testing and refinement of text prompts for Generative AI tools to support business use cases at Devasthali Infotech Private Limited.

We appreciate the candidate's contribution and wish them success in their future academic and professional endeavors.

Date of Issue: 17-09-2026

Place: Noida

Authorized Signatory

Name: Rishi Kumar

DESIGNATION: Accounts Manager

Company Name: Devasthali Infotech Private Limited



Devasthali Infotech Private Limited

Corporate Office - A - 42/5, Main Road, Sector - 62, Noida - 201 309 Tel. : 91-120 - 4981445

Table of Contents

1. Executive Summary
 2. Introduction
 3. Organization and Internship Context
 4. Objectives of the Internship
 5. Technical Background: Generative AI and Prompt Engineering
 6. Work Performed During the Internship
 7. Prompt Development and Refinement Methodology
 8. Testing and Evaluation of Prompts
 9. Illustrative Prompt Engineering Examples
 10. Skills and Learning Outcomes
 11. Challenges and Problem-Solving Approach
 12. Professional and Business Relevance
 13. Conclusion
 14. References
- Internship Certificate

1. Executive Summary

This report presents the internship experience of Prajwal Pratap Singh at Devasthali Infotech Private Limited, Noida, undertaken from 15 June 2026 to 15 September 2026. According to the internship certificate, the central area of contribution was the exploration, testing, and refinement of text prompts for Generative AI tools to support business use cases.

The internship focused on an increasingly important layer of Generative AI applications: communicating requirements to AI systems in a way that produces useful, relevant, and consistent outputs. Prompt engineering is not simply about writing longer instructions. It involves defining a task clearly, supplying appropriate context, specifying constraints, choosing an output structure, testing results, identifying failure modes, and iterating until the prompt performs reliably for the intended use.

The report therefore examines the internship work through four connected activities: understanding the business requirement, constructing an initial prompt, testing the generated output, and refining the prompt based on observed weaknesses. This cycle provides a practical framework for improving AI-assisted workflows while keeping the business objective at the center of the process.

2. Introduction

Generative Artificial Intelligence has changed how organizations approach tasks involving text generation, summarization, classification, ideation, information transformation, and other language-based workflows. However, access to a capable AI model does not automatically guarantee a useful business result. The quality of the instruction, the context supplied to the model, and the constraints placed on the output can significantly affect the usefulness of the response.

Prompt engineering addresses this interaction problem. A well-designed prompt translates a human or organizational requirement into instructions that an AI system can interpret consistently. In a business setting, prompt quality must be evaluated not only by whether an answer sounds fluent, but also by whether it follows requirements, avoids unsupported assumptions, uses the correct format, and remains useful across repeated inputs.

The internship provided practical exposure to this process through exploration, testing, and refinement of text prompts for Generative AI tools. The certificate specifically identifies these activities as the contribution made during the internship.

3. Organization and Internship Context

The internship was completed at Devasthali Infotech Private Limited. The certificate identifies the company's corporate office as A-42/5, Main Road, Sector-62, Noida-201309.

The internship context described in the certificate is centered on Generative AI prompt work. Rather than assigning unsupported details about the company's wider operations, this report limits its organization-specific discussion to the information documented in the certificate and focuses technically on the internship's stated area of work.

Organization	Devasthali Infotech Private Limited
Location	A-42/5, Main Road, Sector-62, Noida-201309
Internship Start	15 June 2026
Internship End	15 September 2026
Internship Area	Generative AI text prompt exploration, testing and refinement
Certificate Issued	17 September 2026
Authorized Signatory	Rishi Kumar, Account Manager

4. Objectives of the Internship

The certificate establishes the broad nature of the internship work. Based on that documented scope, the following objectives describe the practical purpose of the work without attributing unsupported company-specific projects:

- Understand how text prompts can be designed for Generative AI tools.
- Translate business requirements into clear and testable AI instructions.
- Explore alternative prompt structures and compare the resulting outputs.
- Test prompts against expected requirements and identify weaknesses.
- Refine prompts to improve clarity, relevance, consistency, and adherence to constraints.
- Understand the importance of output formatting when AI-generated content is used in a business workflow.
- Develop a systematic approach to iteration rather than relying on one-shot prompting.

5. Technical Background: Generative AI and Prompt Engineering

5.1 Generative AI

Generative AI systems are designed to produce new content from user-provided instructions and context. For language-oriented systems, the output may include explanations, summaries, drafts, structured information, classifications, or other forms of text. The model does not simply retrieve a single fixed answer; it generates an output based on patterns learned during training and the information contained in the current interaction.

For business use, this flexibility is useful but also creates a practical challenge: an instruction that is vague or underspecified can lead to outputs that are plausible but unsuitable. Prompt engineering is one method of reducing that gap.

5.2 Prompt Engineering

Prompt engineering is the process of designing and iteratively improving instructions given to a Generative AI model. A useful prompt generally makes five elements clear: the task, the context, the constraints, the expected output, and any quality criteria.

Task	What the AI is required to do.
Context	Information the AI needs to understand the task.
Constraints	Rules such as length, scope, tone, exclusions, or required fields.
Output Format	The structure in which the response should be returned.
Quality Criteria	Conditions that define whether the result is acceptable.

The important point is that prompt engineering is iterative. A first prompt is a hypothesis about how the requirements should be communicated. Testing reveals where that hypothesis fails, and refinement converts those observations into a better instruction.

5.3 Common Prompt Components

Role or perspective: establishes the type of response expected from the system.

Objective: states the actual task instead of relying on implied intent.

Context: provides relevant background, data, definitions, or examples.

Constraints: prevents unwanted scope expansion and specifies boundaries.

Output schema: defines headings, bullets, tables, JSON fields, or other required structures.

Examples: demonstrates the expected pattern when the task benefits from few-shot guidance.

Validation criteria: tells the evaluator what properties make an answer acceptable.

6. Work Performed During the Internship

The internship certificate specifically states that the work involved the exploration, testing, and refinement of text prompts for Generative AI tools to support business use cases. The work can therefore be represented as a practical prompt-development cycle.

6.1 Exploration

Exploration involves examining how different instructions affect the generated output. Instead of assuming that the first wording is optimal, multiple formulations can be considered. Differences may involve the level of detail, order of instructions, explicit constraints, requested format, or inclusion of examples.

6.2 Testing

Testing involves running prompts and examining the resulting outputs against the intended

requirement. The purpose is not merely to decide whether the answer sounds good. A useful test checks whether the response is relevant, complete, correctly formatted, appropriately scoped, and consistent with the instructions.

6.3 Refinement

Refinement means modifying the prompt based on observed weaknesses. For example, if an output is too broad, the prompt can narrow the scope. If important information is omitted, required fields can be made explicit. If formatting varies between outputs, an output schema or stronger formatting instruction can be introduced.

6.4 Business-Oriented Prompt Design

A business prompt should be designed around the desired outcome rather than around the novelty of the AI technology. The relevant question is not 'What can the model generate?' but 'What useful task should the model perform, under what constraints, and how will the result be checked?'

7. Prompt Development and Refinement Methodology

A repeatable methodology makes prompt work more reliable. The following workflow is suitable for business-oriented prompt development and reflects the exploration-testing-refinement cycle documented for the internship.

1. Define the business requirement in one clear sentence.
2. Identify the information the model must receive to perform the task.
3. Write a baseline prompt with explicit instructions and constraints.
4. Define what a successful output should contain and what it must avoid.
5. Run the prompt using representative inputs.
6. Compare the output with the acceptance criteria.
7. Record failure patterns such as missing information, excessive verbosity, incorrect format, or unsupported assumptions.
8. Modify only the relevant part of the prompt where possible so that the effect of each change can be understood.
9. Retest the revised prompt using more than one representative input.
10. Document the final prompt, its intended use, and the conditions under which it performs reliably.

7.1 Prompt Iteration Example

The following is an illustrative example of the refinement process, not a claim about a specific client task performed during the internship.

Initial instruction: 'Summarize this customer feedback.'

Problem: The instruction does not define the desired length, categories, or output structure. Different responses could emphasize different details.

Refined instruction: 'Summarize the customer feedback in 5 bullet points. Identify the three most frequent complaints, two positive themes, and any specific product issues mentioned. Do not

invent information that is not present in the feedback.'

The refined version is stronger because it defines the task, output structure, scope, and a constraint against unsupported information. This illustrates the type of reasoning used in prompt refinement.

8. Testing and Evaluation of Prompts

Prompt testing should be treated as an evaluation activity rather than an informal 'looks good' check. A business prompt can produce fluent text while still failing the actual requirement. Evaluation should therefore use explicit criteria.

Relevance	Does the output directly address the requested task?
Completeness	Are all required elements present?
Instruction Following	Does the output obey the stated rules and constraints?
Consistency	Does the prompt behave acceptably across representative inputs?
Format Compliance	Does the output follow the requested structure?
Grounding	Does the output avoid unsupported claims where source information is limited?
Usability	Can the result be used in the intended business workflow with minimal correction?

8.1 Common Failure Modes

Vague task definitions leading to **bad** or inconsistent responses.

Missing context causing the model to fill gaps with assumptions.

Conflicting instructions producing unpredictable priorities.

Overly long prompts where important instructions are buried or unclear.

Insufficient output constraints causing inconsistent formatting.

No explicit handling of missing information.

Testing on only one example, which can hide weaknesses that appear on other inputs.

9. Illustrative Prompt Engineering Examples

The examples in this section are technical illustrations created to explain prompt-engineering principles. They are not presented as documented company projects.

Example 1 – Structured Business Summary

Weak prompt: 'Summarize the report.'

Improved prompt: 'Summarize the supplied report in four sections: Executive Summary, Key Findings, Risks, and Recommended Next Steps. Use only information present in the report. If a section has no supporting information, state that explicitly rather than inventing details.'

Example 2 – Classification

Weak prompt: 'Classify these customer messages.'

Improved prompt: 'Classify each message into exactly one of these categories: Billing, Technical Issue, Account Access, Product Feedback, or Other. Return a table with Message ID, Category, and one-sentence Reason. If the message cannot be classified confidently, use Other.'

Example 3 – Content Drafting

Weak prompt: 'Write an email about the product.'

Improved prompt: 'Write a professional customer email introducing the product. Keep it between 120 and 150 words. Include the product's primary benefit, one concrete use case, and a clear call to action. Do not make claims about pricing, performance, or features unless those details are supplied in the input.'

10. Skills and Learning Outcomes

The internship provided exposure to a workflow that combines communication, technical reasoning, testing, and iterative improvement. The following learning outcomes are directly aligned with the documented internship activity while remaining general rather than claiming unsupported project-specific achievements.

- Understanding of the basic principles of Generative AI interaction.
- Ability to convert an ambiguous requirement into clearer instructions.
- Improved awareness of the relationship between prompt structure and output quality.
- Experience with iterative testing and refinement as a problem-solving method.
- Better understanding of output constraints and structured responses.
- Recognition of hallucination and unsupported-assumption risks.
- Ability to think about AI outputs from a businessability perspective.
- Improved technical documentation and communication of prompt requirements.

11. Challenges and Problem-Solving Approach

Prompt engineering presents a different type of debugging problem from conventional programming. In traditional software, a defect is often associated with a specific line or component. With Generative AI, the same prompt can produce different outputs, and a failure may result from ambiguity, insufficient context, conflicting instructions, or an unsuitable evaluation criterion.

11.1 Ambiguity

When a prompt leaves important decisions unstated, the model may make those decisions itself. The practical response is to identify which decisions should be controlled by the prompt and specify them explicitly.

11.2 Inconsistent Formatting

If outputs vary in structure, the prompt should define the required format more precisely. Examples, field lists, headings, tables, or machine-readable schemas can reduce unnecessary variation.

11.3 Unsupported Information

A fluent answer is not automatically a factual answer. Where source material is limited, prompts should explicitly instruct the system to distinguish provided information from missing information and avoid inventing details.

11.4 Over-Engineering

Adding more instructions does not automatically make a prompt better. Excessive or contradictory instructions can make the task harder to follow. A good refinement changes the prompt to address an observed failure rather than adding rules without a clear purpose.

12. Professional and Business Relevance

The business value of prompt engineering comes from converting general-purpose AI capabilities into repeatable workflows. A prompt becomes more valuable when it can be reused, evaluated, documented, and maintained instead of being treated as a one-time chat instruction.

- Standardizing recurring text-generation tasks.

- Improving consistency in AI-assisted workflows.

- Reducing manual rewriting when outputs follow a defined structure.

- Making AI-assisted processes easier to test and document.

- Creating clearer interfaces between human requirements and AI systems.

- Supporting experimentation while maintaining explicit quality criteria.

At the same time, prompt engineering should not be treated as a substitute for verification. Business-critical outputs may require human review, source validation, access controls, privacy safeguards, and domain-specific checks. A prompt can reduce certain failure modes, but it cannot guarantee correctness.

13. Conclusion

The internship at Devasthali Infotech Private Limited, completed from 15 June 2026 to 15 September 2026, focused on exploring, testing, and refining text prompts for Generative AI tools in support of business use cases. This scope is explicitly documented in the internship certificate.

The key practical lesson from this area of work is that effective AI interaction is an iterative engineering process. A useful prompt begins with a clearly defined objective, provides the

necessary context, establishes constraints, specifies the desired output, and is tested against explicit quality criteria. Refinements should be driven by observed failures rather than by adding complexity for its own sake.

The internship therefore provided a practical foundation in Generative AI development and in the disciplined process of testing and improving AI instructions for business-oriented use.

14. References

Devasthali Infotech Private Limited, Internship Certificate issued to Prajwal Pratap Singh, dated 17 September 2026.

General technical concepts of Generative AI and prompt engineering used for explanatory sections of this report.

Note: No external company profile, client information, proprietary project details, or specific AI platform usage has been added because such information is not contained in the supplied internship certificate.